

Homework 2: Source-Monitoring Model (TreeBUGS)

Daniel W. Heck

2025-03-14

The following exercises serve as practice tasks for TreeBUGS and are based on the supplementary material of the paper Heck, Arnold, & Arnold (2018) in *Behavior Research Methods*.

Data

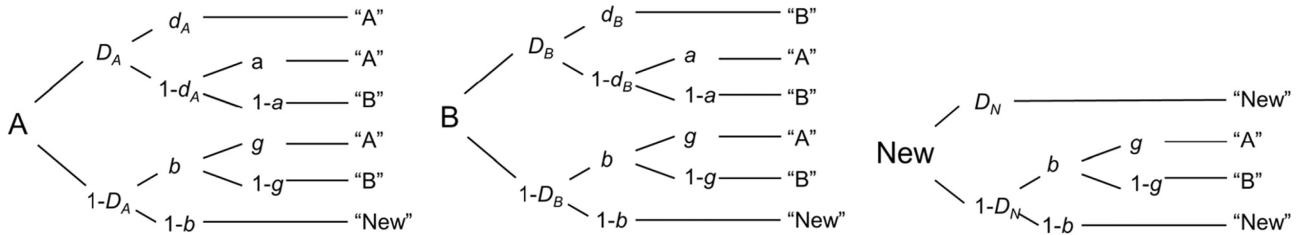
The data are taken from Arnold et al. (2013) from a study on source memory. After learning words from two sources (E = expected and U = unexpected), participants categorized studied and new words in a test list as “old - source E”, “old - source U” or “new (N)”. This results in 3×3 response categories: (EE,EU,EN), (UE,UU,UN), and (NE,NU,NN). Moreover, participants were assigned to two between-subject groups that differed in the time of schema activation (“retrieval” vs. “encoding”).

All data are in the folder `/data`:

- Response frequencies are in `data_both.csv` (and also split by group in `data_retrieval.csv`, `data_encoding.csv`).
- The data files `pc_***.csv` contain the continuous covariate “perceived contingency”, a hypothesized predictor for the response bias parameter for source guessing (parameter `b`).
- The data file `group.csv` contains a discrete grouping variable for the data set `data_both.csv`

MPT Model

The MPT model equations for the 2-High-Threshold Source-Monitoring Model (2HTSM) and the corresponding parameter constraints are in the folder `/model` (files: `2htsm.eqn` and `restrictions.txt`). The model assumes that items are recognized as OLD or NEW with probability $D_1 = D_2 = D_3$ (item memory). Moreover, conditional on recognizing an item as OLD or NEW, a person remembers the source correctly with probability $d_1 = d_2$ (source memory). Finally, the model assumes a source-guessing parameter $a = g$ and one old/new-guessing parameter b .



The first lines of the EQN file for expected items (source E) are:

```
E      EE      D1*d1          # item detection and source detection
E      EE      D1*(1-d1)*a    # item detection and source guessing
E      EU      D1*(1-d1)*(1-a)
E      EE      (1-D1)*b*g      # guessing for both item and source
E      EU      (1-D1)*b*(1-g)
E      EN      (1-D1)*(1-b)
```

Exercises

1. Plot the individual frequencies in the “retrieval” condition to assess heterogeneity.
2. Fit a latent-trait MPT model to the data from the “retrieval” condition.
3. Check MCMC convergence and extend the sampling if necessary.
4. Test the model fit graphically and with posterior-predictive p -values.
5. Plot the parameter estimates.
6. *Within-subject test:*
 - Compute and summarize the differences in the parameters D (item memory) and d (source memory) using the function `transformedParameters()`.
 - Note that this comparison is not at all substantively meaningful!
7. *Between-subject test:*
 - Fit the latent-trait MPT model separately for the data from the “retrieval” and from the “encoding” condition.
 - Based on the two fitted models, compute the difference of the parameter D across conditions using the function `betweenSubjectMPT()`.

Further Exercises (not part of the homework!)

If you want to practice more, you may try out the following exercises:

8. *Continuous predictor:* Fit the latent-trait MPT to the “retrieval” condition with the variable “pc” (perceived contingency) as a continuous predictor for the response-bias parameters b and a .
9. *Discrete predictor* (= between-subject tests with identical covariance matrix across groups): Fit the latent-trait MPT to the data from both conditions jointly with the grouping variable “group” (in the data file: `group.csv`) as a discrete, fixed-effects predictor for all 4 parameters.
10. Plot the default prior TreeBUGS uses for the latent-trait MPT.
11. Generate frequencies from the prior predictive distribution for 32 items per MPT tree (i.e., 32 source E, 32 source U, 32 new) and $N = 100$ participants.
 - [Difficult] If you have experience with data manipulation in R: compute the average hit-rate (E or U responses to E or U items) and false-alarm rate (E or U responses to N items) for each replication (in a `for` loop) and plot these rates as points in the ROC space (x = false alarm, y = hit rate).